Author completion checklist

The remaining decisions and files needed to complete the protocol package, with a common reference for every red field.

Working document. Red **TO COMPLETE** fields identify missing or conflicting author inputs. Complete the fields affecting a procedure before using it at the bench. Recorded settings are not a claim of experimental optimization.

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01 Use the document package

This package brings the assay instructions, preparation guides and records together. Red TO COMPLETE fields identify exact information the method authors still need to supply. Close a field with a verified recipe, file or decision and record the supporting run; a general assurance that a step works does not replace those details.

Document	Use it for
Laboratory protocol	Staining, pairing, stabilization, sorting, custom capture, targeted recovery and sequencing.
Library preparation	Construct verification, cell-library generation, qualification and banking.
Materials and primers	Reagent identities, access requirements, recorded oligos and construct compatibility.
Controls and troubleshooting	Evidence needed at each checkpoint and diagnostic records when a step fails.
Loading and pooling	Sample roles, separate counting units and the information needed to complete validated calculations.
Analysis and worked example	Processing requirements, reporting tables and the files needed for a reproducible example.
Experiment plan	The separate printable run worksheet for recording samples, controls, handoffs and observations.

Document status
The document date identifies this assembled working package. It is not the experimental protocol version, a method DOI or evidence of independent reproduction. The authors must assign and approve those release records.

02 Materials and preparation decisions

Use the same completion IDs across the PDFs and laboratory records. Retain the original decision, its evidence and the date each field was closed. [D1 §12]

TO COMPLETE - C01
Supply exact cell derivatives, accessible materials, full annotated construct sequences, culture conditions and qualification records. Repair and resequence the HLA-B*51:01 and HLA-C*01:02 stocks flagged in the supplied preparation notes.

TO COMPLETE - C02
Complete cloning, packaging and transduction instructions: all reagent identities, formulations, functional-titer basis and selection timing. If TCRAFT is offered, resolve its local-backbone adaptation separately.

TO COMPLETE - C03
Identify the violet tracer, stock molarity, solvent/reconstitution, final working concentration, staining density, centrifugation and permitted holds.

TO COMPLETE - C04
Specify pairing medium volumes, total volume, cell density, temperature/CO2, vessel and shaker geometry, and the timing and scale supported by reference runs.

03 Capture and library decisions

TO COMPLETE - C05
Resolve the BS3 temperature conflict (37 °C versus room temperature). Complete the product/stock formulation, PBS specification, cell load, absolute glycine concentration and pH, rotation, washes and hold limits.

TO COMPLETE - C06
Provide the sorter model, pressure/rate/mode, filters, controls and gating files, collection/concentration method, intact-pair assessment and measured post-sort recovery.

TO COMPLETE - C07
Reconcile individual-cell versus intact-pair counts and provide the validated loading worksheet. Specify the custom 10x capture modification, complete RT/cleanup instructions and annotated construct/read architecture.

TO COMPLETE - C08
Resolve CZP31 versus CZP31.0, the supported TCR forward primer, and indexed-primer aliases. Verify binding sites and orientation against every released construct.

TO COMPLETE - C09
Supply separate complete recipes for the intended PCR1 input types and all PCR2 branches: units, stocks, template allocation, cycling, cleanup and confirmed pause points.

TO COMPLETE - C10
Specify construct-dependent fragment QC, cleanup procedures and the pooling calculation basis. Provide the missing calculator, total and per-branch volume constraints, and a worked calculation.

TO COMPLETE - C11
Pin the sequencing kit/manual revision, platform and flow cell, input/loading recipe, QC, acquisition/basecalling settings and run stopping criteria.

04 Analysis, reference evidence and release

TO COMPLETE - C12
Provide executable read processing and analysis with a pinned environment, references, barcode/index architecture, assignment and ambiguity rules, background/normalization/calling definitions and a shareable worked example with expected outputs.

TO COMPLETE - C13
Name the validated controls, define replicate design and reference-run acceptance criteria, and supply the supporting gates, traces and output tables. Record reproduction by an operator outside the originating workflow when it has occurred.

TO COMPLETE - C14
Supply method authors/contact, final protocol version and persistent identifier, material access routes, separate text/code/data/material reuse terms, permitted example data, change history and checksums.

Release record	Author entry
Method authors and contact	TO COMPLETE: names, affiliation and support route.
Protocol version and date	TO COMPLETE: experimental version and approval date.
Method paper	Parallel mapping of TCR-pMHC interactions decodes T cell recognition. TO COMPLETE: approved citation and persistent link.
Document/code/data licenses	TO COMPLETE: separate reuse terms for each artifact class.
Material distribution	TO COMPLETE: repository/accession or distribution route and applicable transfer terms.
Reference example	TO COMPLETE: permitted dataset, source run, expected outputs and archive link.
Change record	TO COMPLETE: changed steps, reason, supporting evidence and affected artifacts.

05 Check the assembled release

1. Resolve each red field against a named file, recipe or validated decision. Update all documents that use that value, construct or identifier.
2. Have a second reader check reagents, units, primer sequences, sample/index assignments and linked resources against the selected laboratory records.
3. Run the analysis example from the documented starting files in the specified environment. Compare assignments, QC tables and figures with the reference outputs using declared tolerances.
4. Have an operator outside the originating workflow use only the package and its declared materials/support. Record each clarification needed and revise the affected instructions.
5. Assign the experimental protocol version, archive the selected vendor revisions and release files, record reuse terms and checksums, and update the website links together.

Keep raw observations, exclusions and analysis decisions separate from model predictions. A shared partition does not alone establish recognition, and an unobserved combination is not a verified non-interaction. [D1 §§8-12]

Source notes

Sources are the method authors' laboratory records, as transcribed in the supplied consolidation. The internal source files are not included in this download.

D1. Consolidated CLASP-seq protocol draft. Author-supplied consolidation of laboratory records, September 11, 2026, sections 8-12. The completion register records unresolved inputs; it does not establish experimental validation.

Protocol and resources